VIETNAM NATIONAL UNIVERSITY

UNIVERSITY OF SCIENCE

Project 2

Building a Serverless Web Application with Zero-Cost & Secure Architecture

Course Code: CSC11007

Student:

23127158 - Nguyen Thanh Gia Bao

23127404 - Le Tuan Loc

23127438 - Dang Truong Nguyen

Academic Supervisors:

Nguyen Thanh Quan

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CHAPTER 1

Frontend Security Evidence

1.1. SE-1: S3 Block Public Access enabled

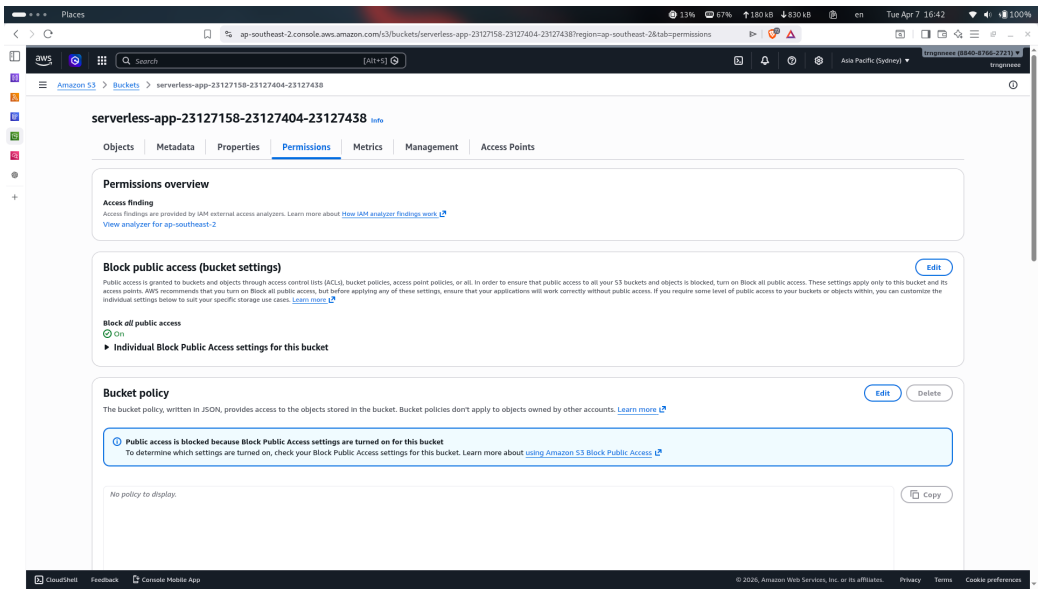

Figure 1.1: SE-1

1.2. SE-2: Direct S3 access rejected

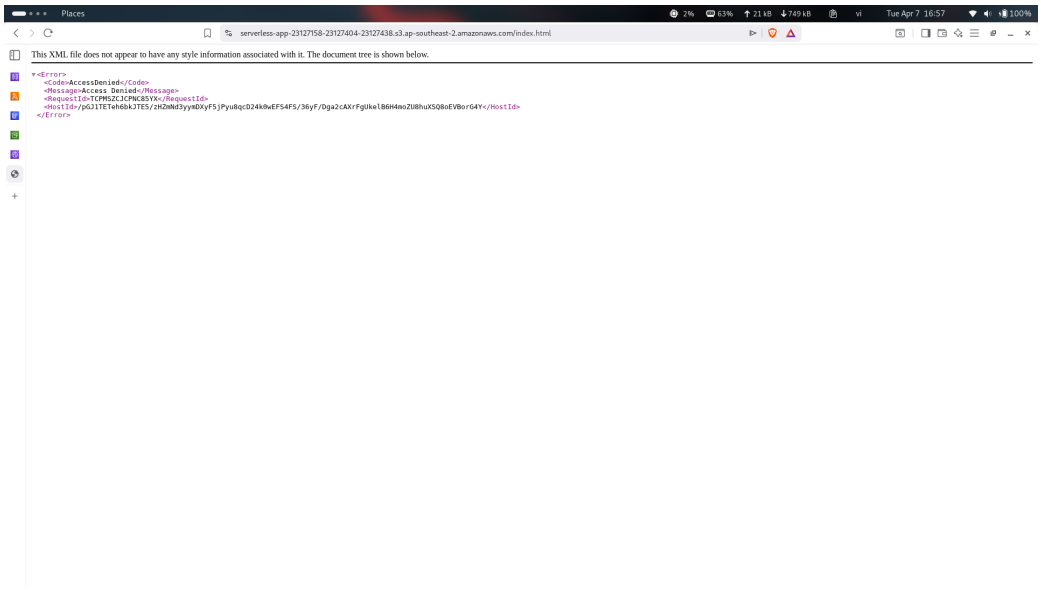

Figure 1.2: SE-2

1.3. SE-3: CloudFront access succeeds

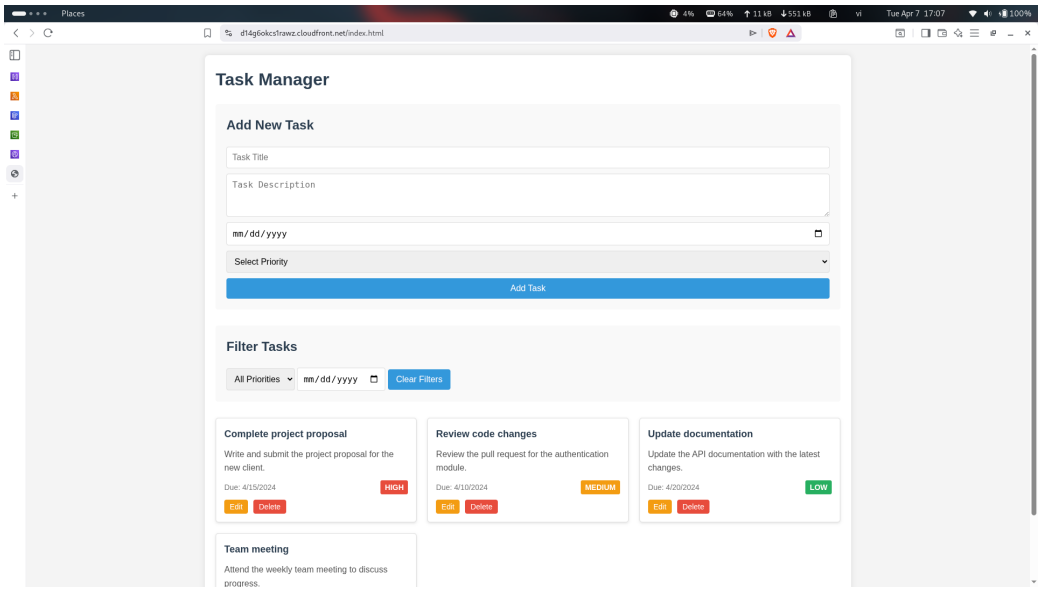

Figure 1.3: SE-3

1.4. SE-4: OAC attached to CloudFront

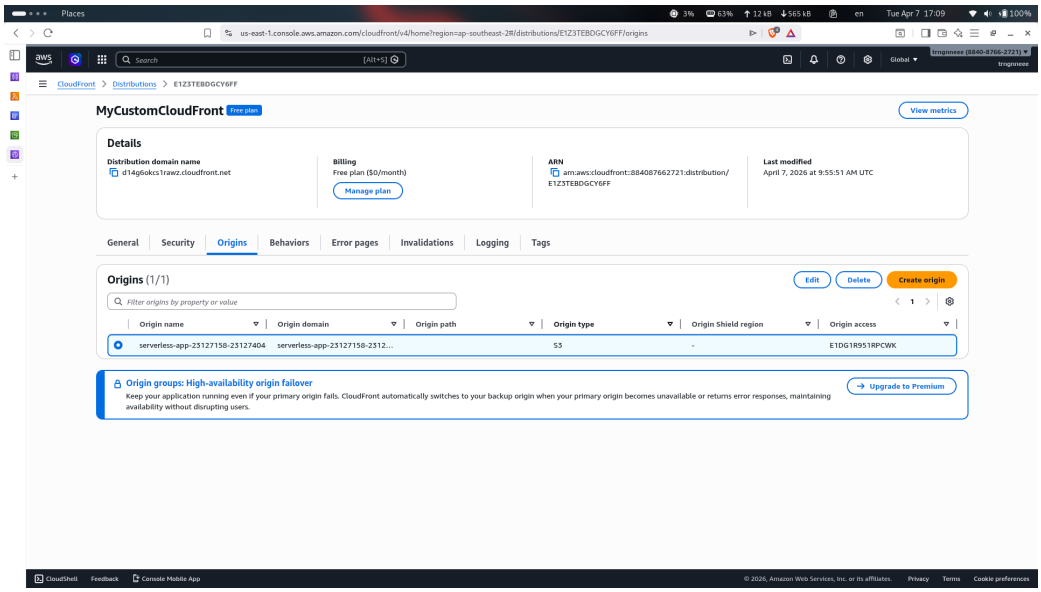

Figure 1.4: Origins Access List

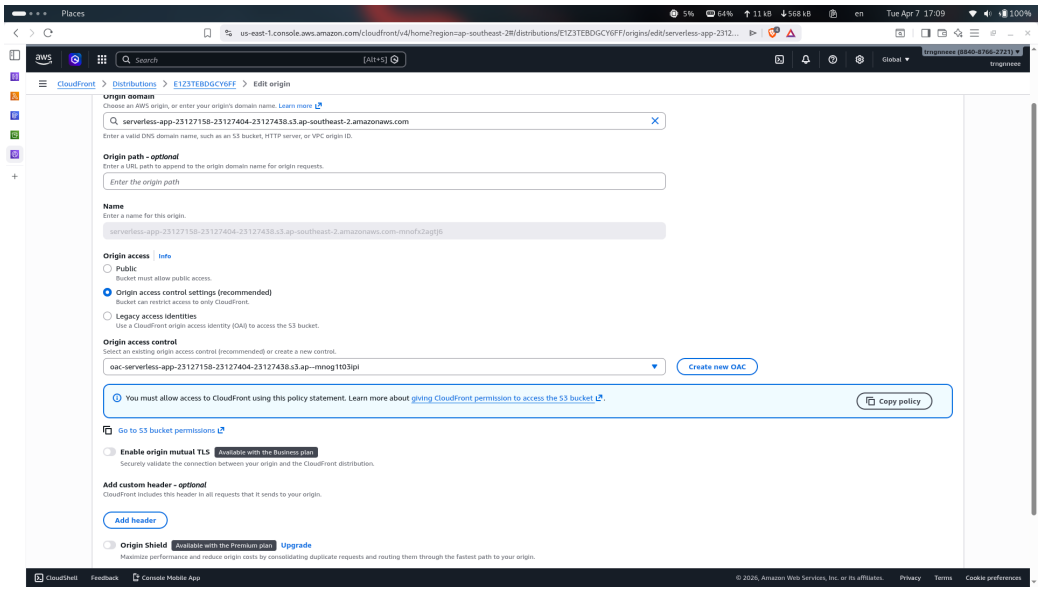

Figure 1.5: Origins Access Setting

CHAPTER 2

Network Evidence

2.1. NE-1: VPC Endpoint created for DynamoDB

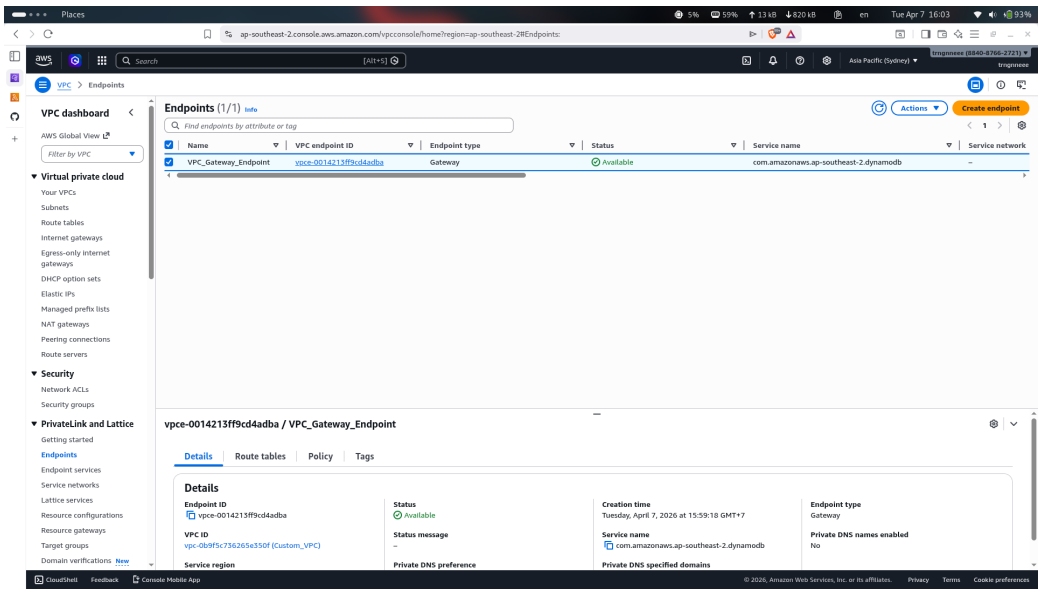

Figure 2.1: NE-1

2.2. NE-2: Lambda has VpcConfig

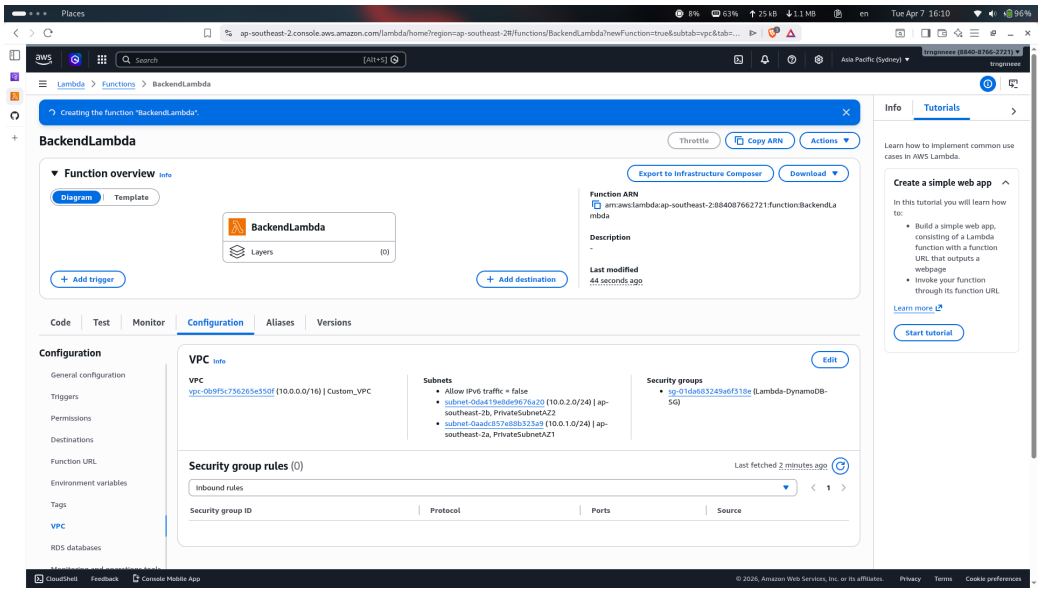

Figure 2.2: NE-2

2.3. NE-3: Route table has Endpoint route

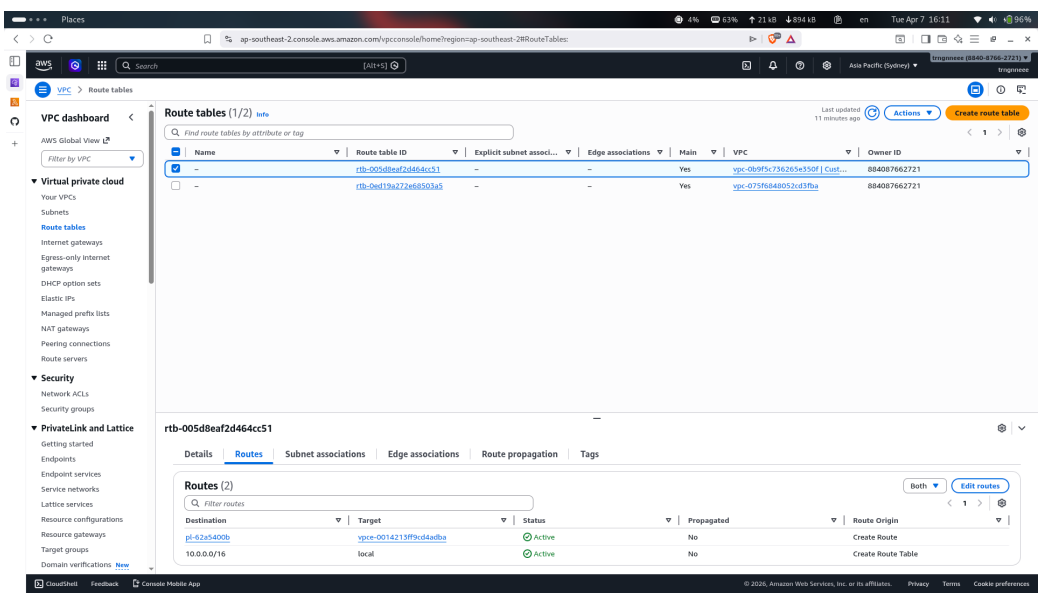

Figure 2.3: NE-3

2.4. NE-4: No NAT Gateway exists

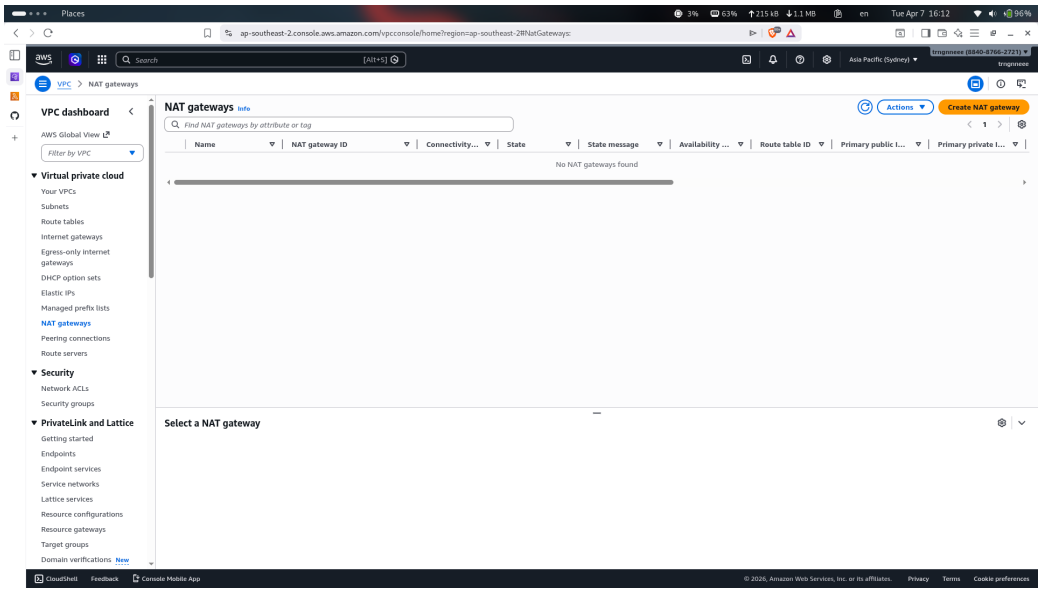

Figure 2.4: NE-4

2.5. NE-5: Lambda successfully calls DynamoDB

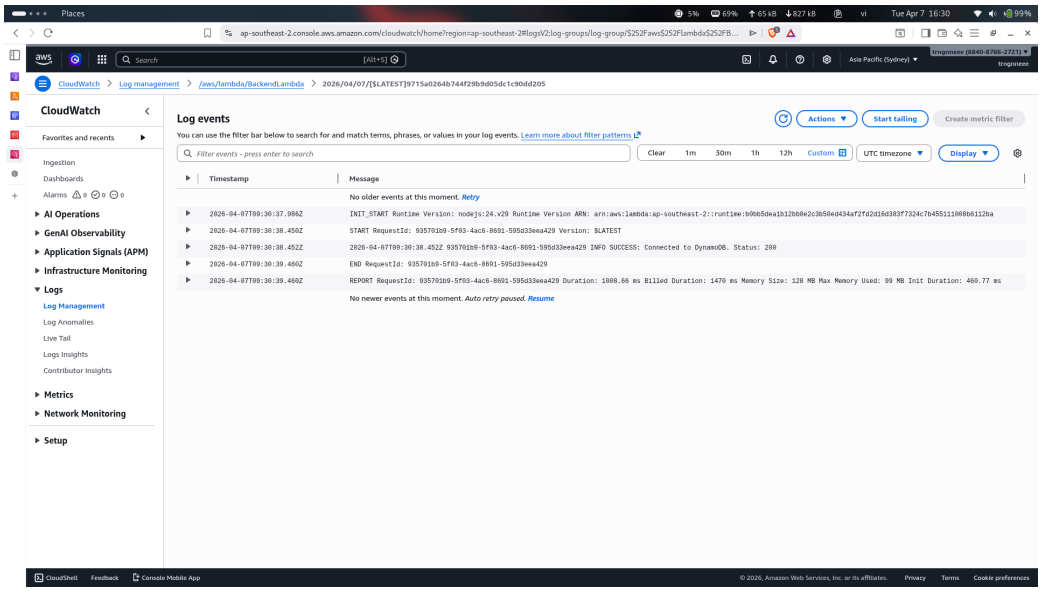

Figure 2.5: NE-5

CHAPTER 3

IAM evidence

3.1. IM-1: Two separate IAM Roles

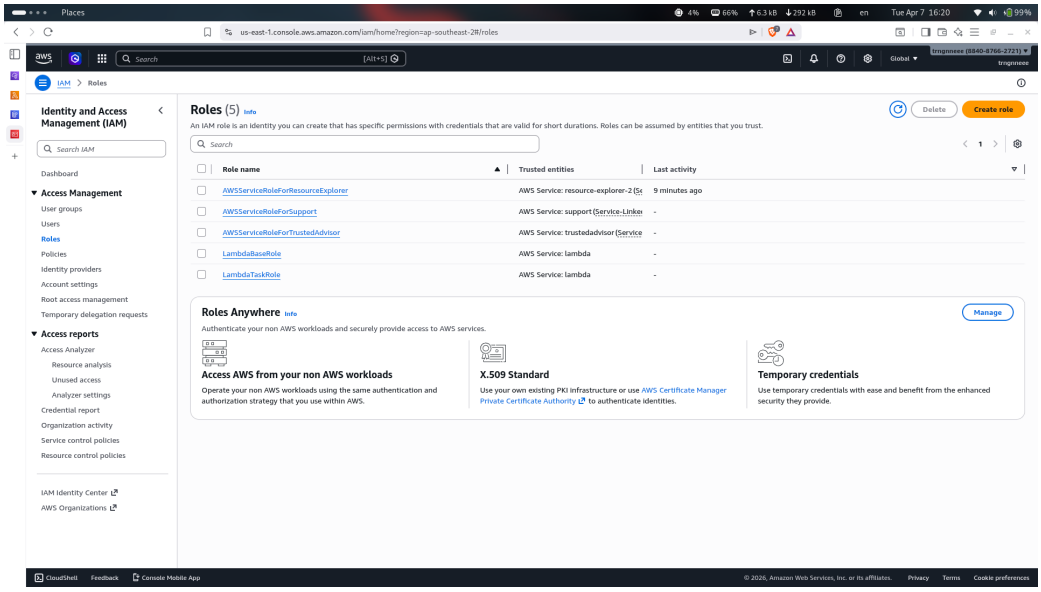

Figure 3.1: IM-1

3.2. IM-2: Policy specifies exact resource ARN

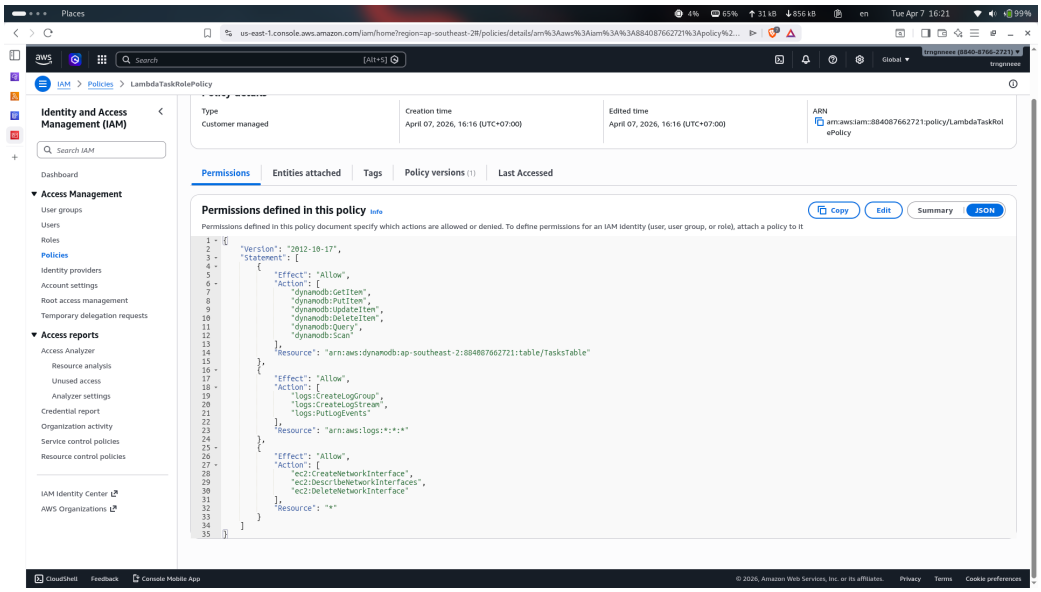

Figure 3.2: Lambda Base Role's Policy

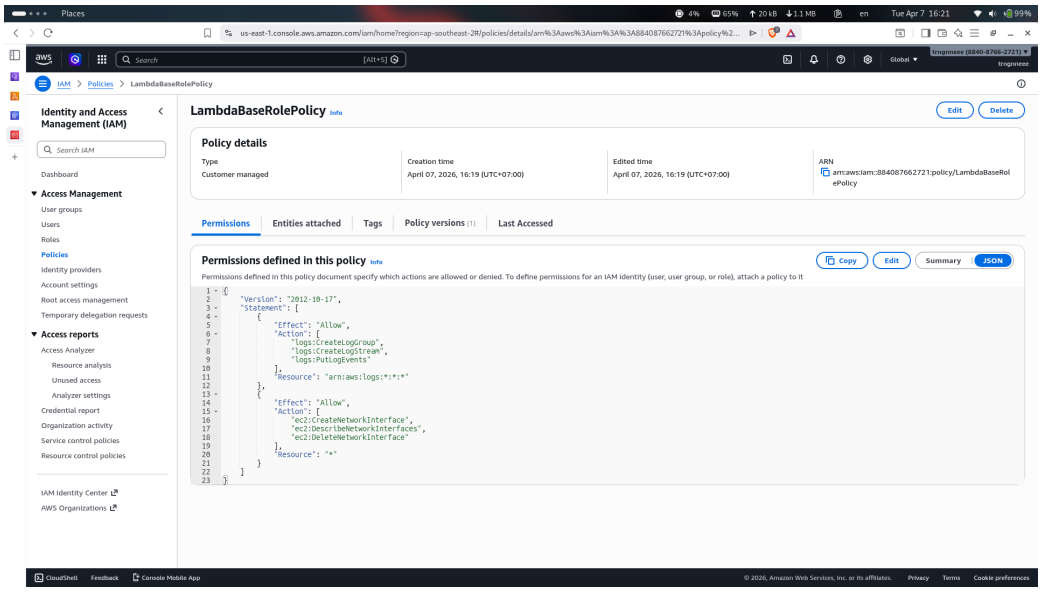

Figure 3.3: Lambda Task Role's Policy

3.3. IM-3: Lambda attached to the correct Role

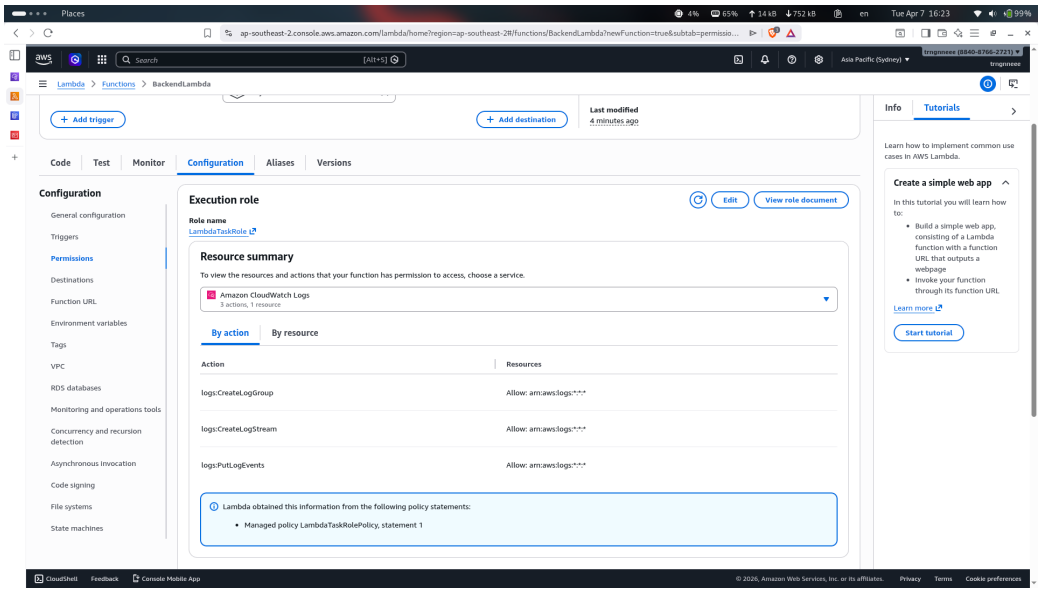

Figure 3.4: Lambda Execution Role by Action

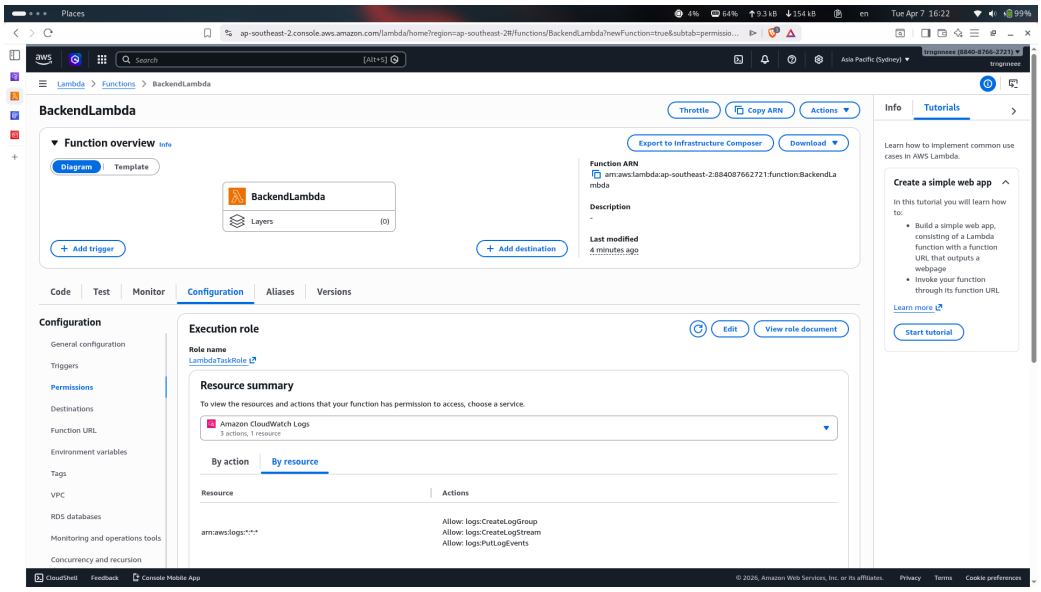

Figure 3.5: Lambda Execution Role by Resources

CHAPTER 4

Log evidence

4.1. Successful request

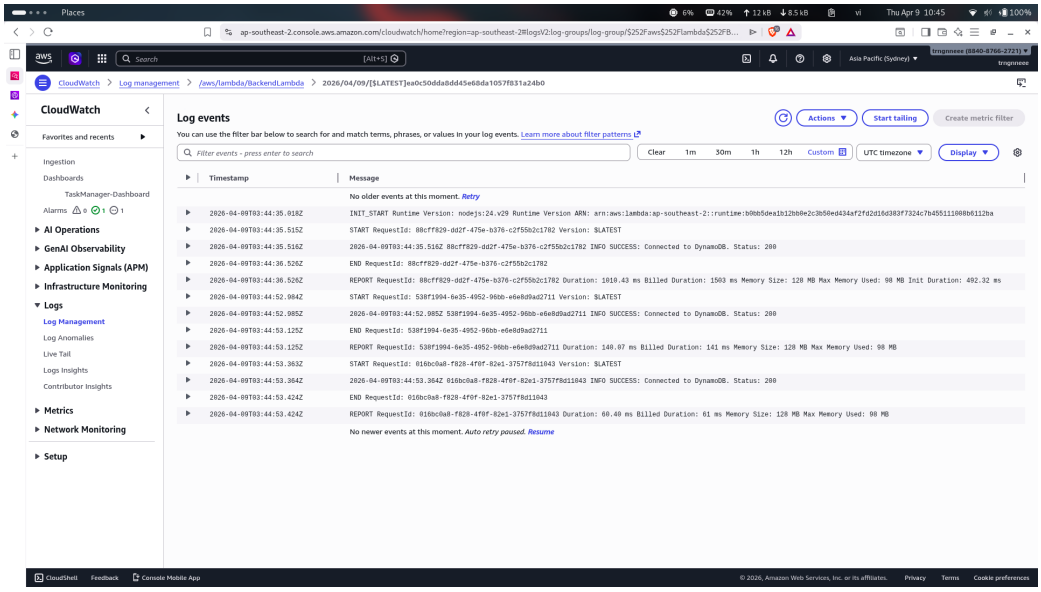

Figure 4.1: Log showing the REPORT line with StatusCode 200, Duration, Billed Duration, and Memory Used

4.2. Error request

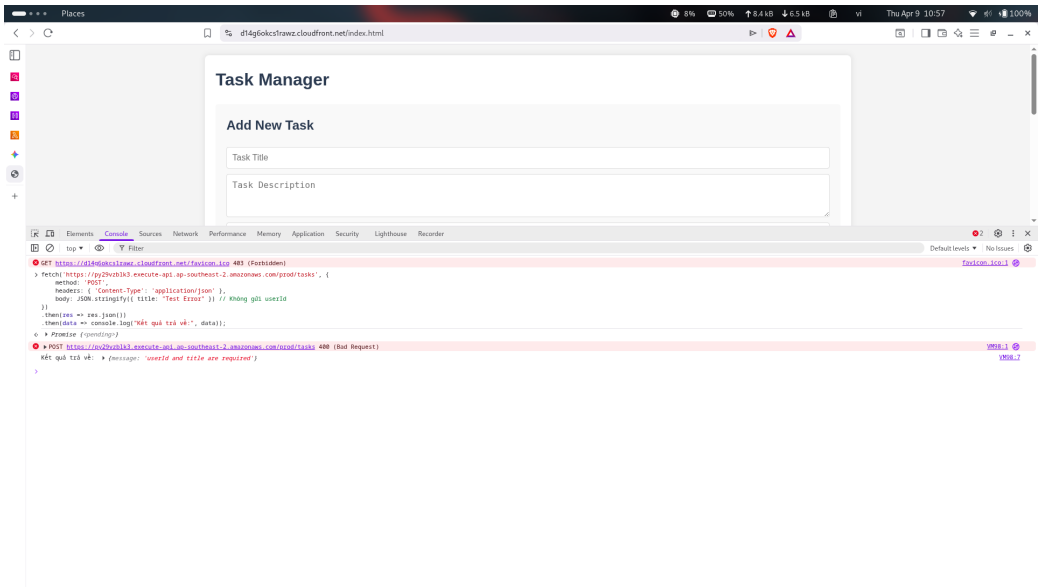

Figure 4.2: Sending missing input

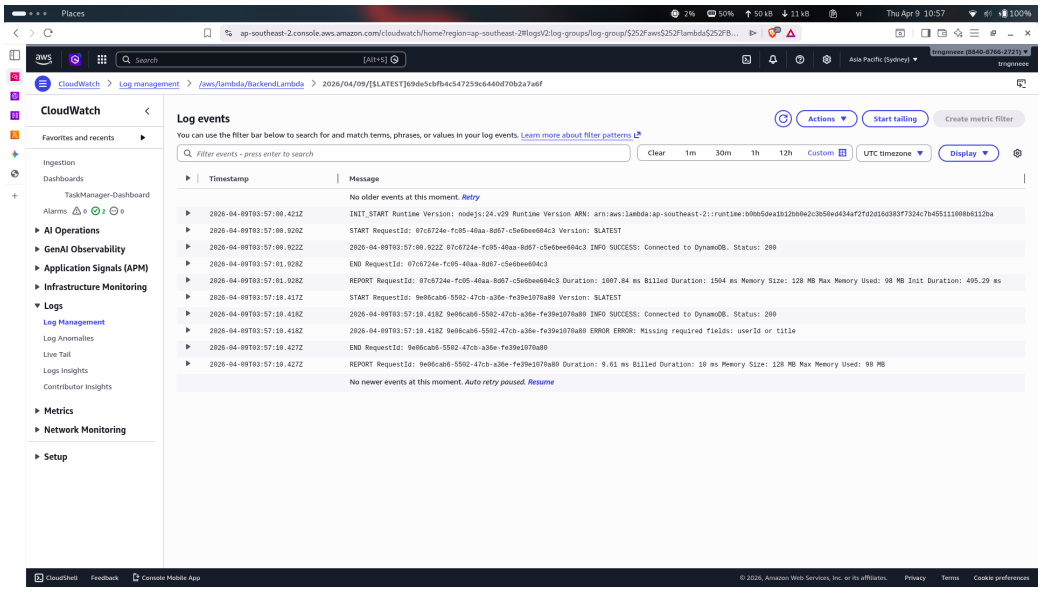

Figure 4.3: Log showing an ERROR entry or stack trace triggered by sending missing input

CHAPTER 5

Monitoring Dashboard

5.1. CloudWatch Dashboard

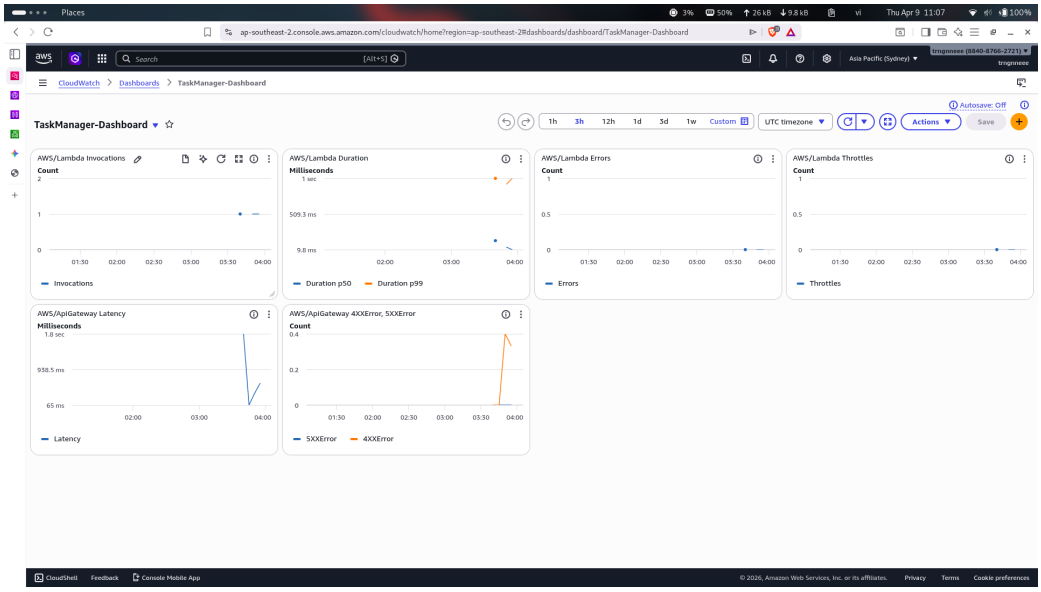

Figure 5.1: CloudWatch Dashboard showing at least 5 widgets with live data

5.2. Alarm

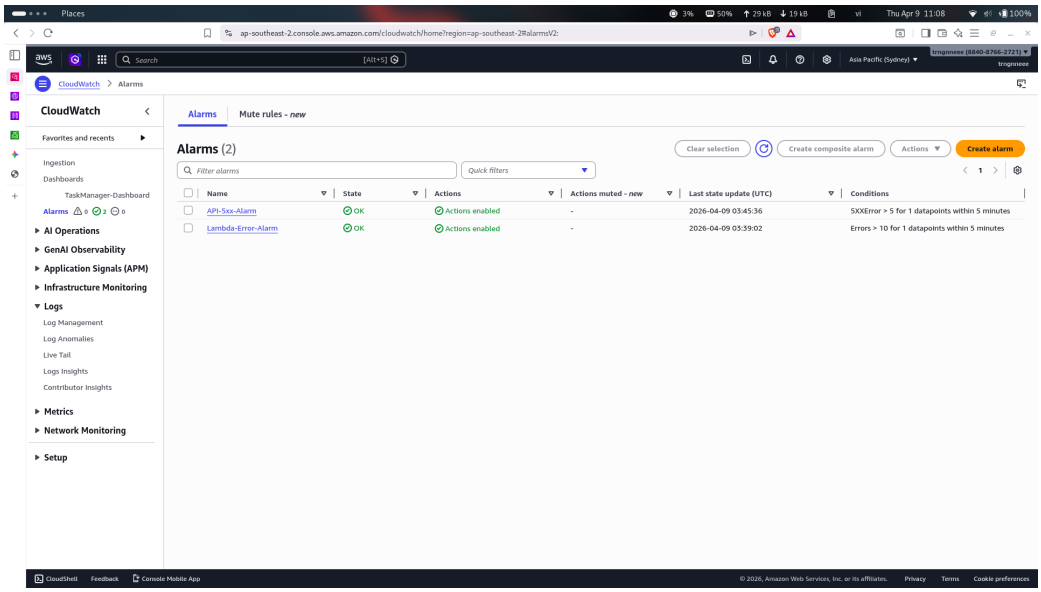

Figure 5.2: 2 alarms created

5.3. AWS Budget configuration page

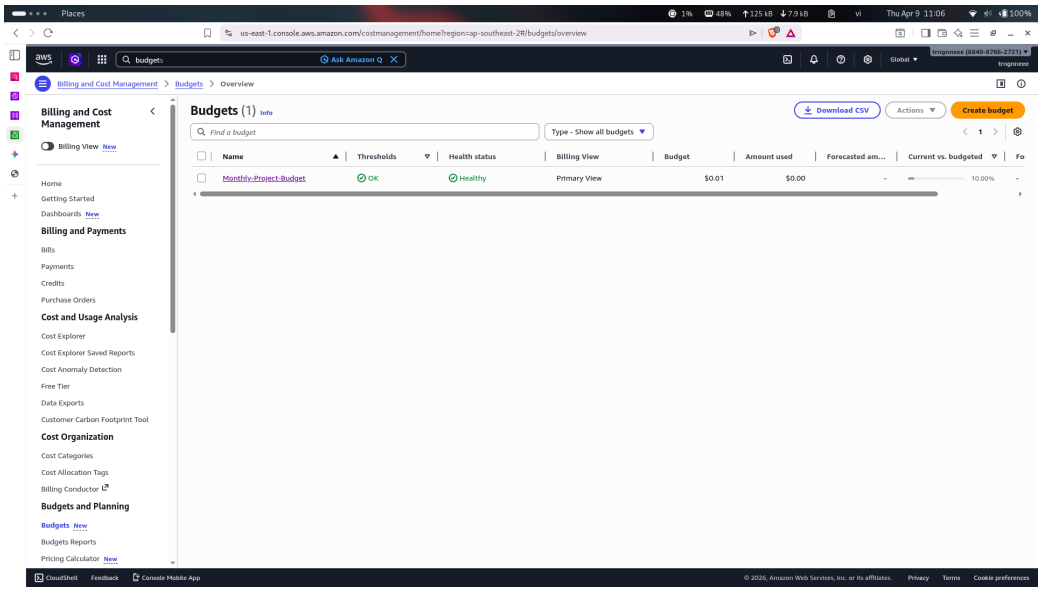

Figure 5.3: AWS Budget configuration page

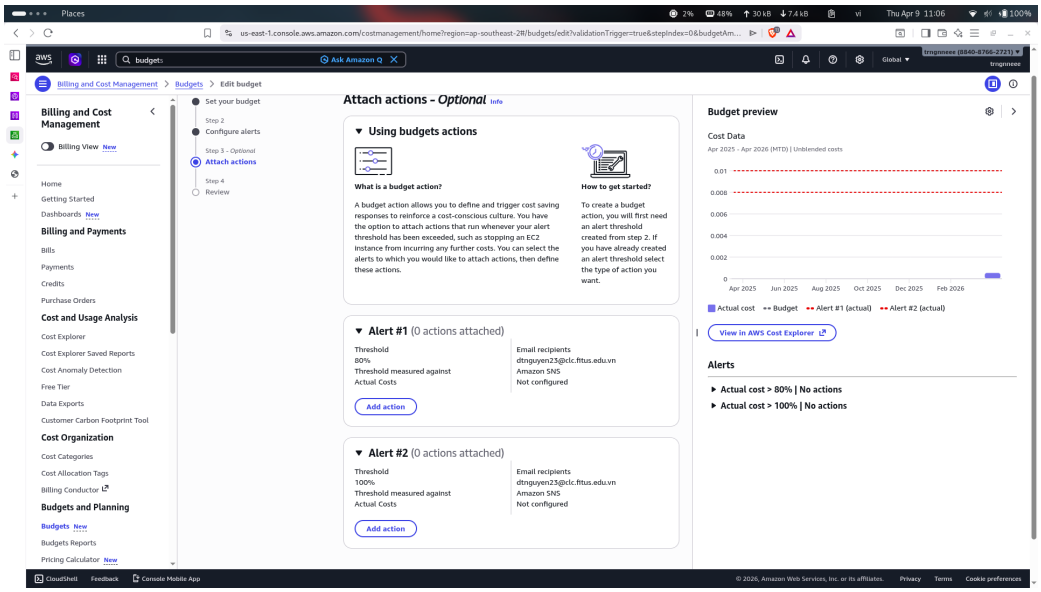

Figure 5.4: AWS Budget alert configuration page

CHAPTER 6

Cost Report

6.1. AWS Cost Explorer/Billing Dashboard

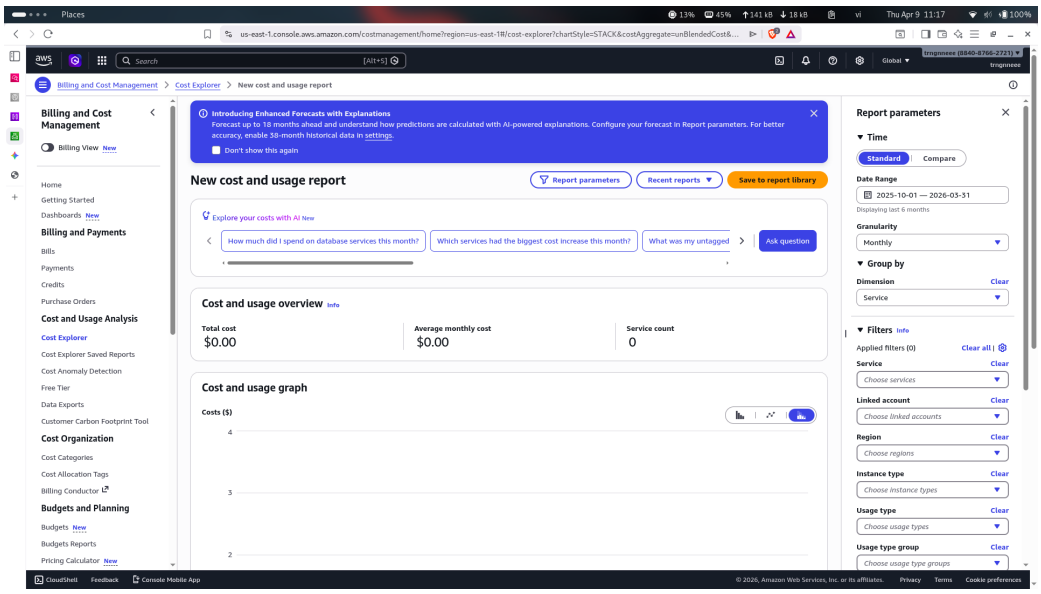

Figure 6.1: AWS Cost Explorer

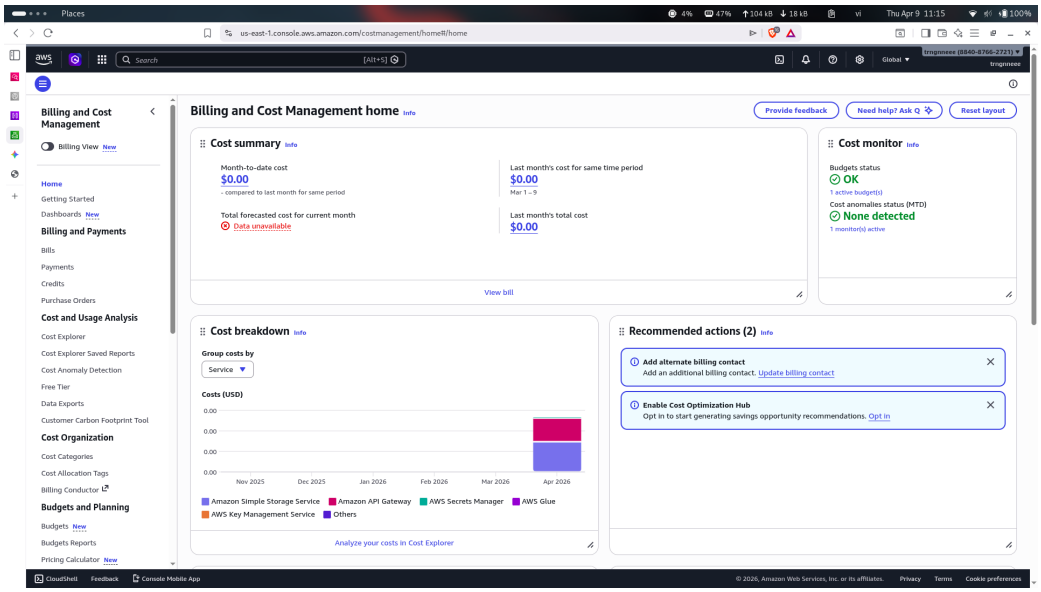

Figure 6.2: Billing Dashboard